

### **Question bank Module 3**

- 1 Explain frequency modulation with neat waveform. Explain phase modulation with neat waveform
- 2 With neat waveform explain phase shift keying
- 3 Analyze how the noise amplitude added to FM introduces a phase shift
- 4 Explain preemphasis and Deemphasis with neat diagram
- 5 Analyze the working of pulse averaging discriminator with neat diagram and waveform
- 6 What is superheterodyne receiver ? Explain the working of superheterodyne receiver with neat block diagram
- 7 Explain the working of varactor modulator with neat circuit diagram
- 8** Explain the advantages of FM over AM
- 9 Define modulation index . Explain the frequency spectrum of FM
- 10 Explain the frequency modulation using crystal oscillator
- 11 Explain the frequency modulation voltage controlled oscillator
- 12 Explain frequency demodulation using slope detector
- 13 Explain quadrature detector with neat diagram